Fiscal Year:	FY 2017	**Task Last Updated:**	FY 08/02/2016
PI Name:	Thaxton, Sherry Ph.D.		
Project Title:	Vehicle NHV and Habitability Assessment		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2013	**End Date:**	02/05/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Williams, Thomas	**Contact Phone:**	281-483-8773
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 2/5/2017 due to PI S. Thaxton's move to Human Factors & Behavioral Performance Deputy Element Scientist. Project continues with Maya Greene, Ph.D. (Ed., 6/26/2017) NOTE: Element change to Human Factors & Behavioral Performance; previously Space Human Factors & Habitability (Ed., 1/18/17) NOTE: End date changed to 6/29/2018 per E. Connell/M. Whitmore (JSC HRP)--Ed., 1/21/16 NOTE: End date changed to 10/1/2016 to accommodate final reporting (Ed., 8/3/15) NOTE: Title change to "Vehicle NHV and Habitability Assessment"; previously "Habitability Assessment of International Space Station" (Ed. 7/14/14) NOTE: Risk/Gaps per E. Connell/HRP (Ed., 3/20/14) NOTE: Start date changed to 10/1/13 (from 6/27/13) per M. Whitmore/JSC (Ed., 2/24/14)		

Key Personnel Changes/Previous PI:	June 2017: Dr. Sherry Thaxton is no longer Principal Investigator as of Feb 6, 2017. New Principal Investigator will be: Maya Greene, Ph.D. (KBRwyle). Project continues with M. Greene as PI. August 2016: Remove Key Personnel Carli Domenico and Alexandra Whitmire; Add key personnel Tom Williams, Jocelyn Dunn. July 2014: Remove co-I: Mihriban Whitmore; Add co-I: Katie Vasser; Add Key Personnel: Ron Archer, Marla Gonzalez. July 2015: Remove Key Personnel: Marla Gonzalez; Add Key Personnel: Rhonda Russo, Carli Domenico.
COI Name (Institution):	Schuh, Susan (MEI Technologies/NASA Johnson Space Center) Vasser, Katie (MEI Technologies/NASA Johnson Space Center) Archer, Ron (Lockheed Martin/NASA Johnson Space Center) Williams, Thomas (Wyle/NASA Johnson Space Center)
Grant/Contract No.:	Directed Research
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The purpose of this study is to assess habitability on the International Space Station (ISS) in order to better prepare for long-duration spaceflight missions of the future. The project deliverables will include information to help prioritize and reduce research gaps, operational inputs to the Human Performance Data Repository, and data for modeling and simulation tool development and validation to use for future designs.The knowledge gained through this project will provide valuable insight into a day-in-the-life of an astronaut as well as providing initial steps to characterize/quantify how we work and live in a microgravity environment during a year-long mission. Thus, it will help address specific research needs identified as part of the Human Research Program's (HRP's) 2012 Habitable Volume Workshop and Standing Review Panel comments, and result in recommendations for future vehicle design layout and minimum net habitable volume (NHV). In addition, another potential outcome of the project will be enhancing the current ISS human factors crew debrief processes, resulting in higher quality data with minimal impact on crew time. This study is led by the Space Human Factors Engineering (SHFE) portfolio under the Space Human Factors and Habitability (SHFH) Element, in collaboration with the Behavioral Health and Performance (BHP) Element. Specific Aims: Specific Aim 1: Characterize the current state of ISS habitability using tools to capture data near real-time. o Aim 1a: Document/quantify details about crew task performance in a long-duration microgravity environment, including influences from the habitable environment and relationship impacts to the behavioral state of crewmembers. o Aim 1b: Determine whether mission duration leads to changes in habitability/human factors reporting by crewmembers. o Aim 1c: Assess the cost versus benefit of implementing near real-time tools compared to traditional post-mission debriefs. Specific Aim 2: Document/characterize details about how crewmembers currently utilize the space on ISS. o Aim 2a: Quantify time spent by crewmembers at workstations/habitation areas. o Aim 2b: Collect naturalistic data to document movement of crewmembers throughout ISS (e.g., frequency of translations between locations). o Aim 2c: Collect evidence to use toward validating estimates of required volume for performance of volume-driving tasks. o Aim 2d: Capture changes in strategies for crew tasks such as translation, stowage handling, etc. throughout the course of the mission.
Rationale for HRP Directed Research:	This research is directed due to a time constraint. This proposal focuses on the research opportunity afforded by the 2015 year-long mission of two crewmembers aboard the International Space Station (ISS).
Research Impact/Earth Benefits:	Innovative technology developed includes the Space Habitability Observation Reporting Tool (iSHORT). iSHORT is an iPad-based application that allows crewmembers to document near real-time observations about their surroundings using text, photographs, video, and audio recordings. iSHORT was reviewed by the Johnson Space Center (JSC) New Technology Evaluation Board, which recommended it for publication in a future edition of the NASA Tech Briefs magazine.
	Data collection has continued for the Vehicle Net Habitable Volume (NHV) and Habitability Assessment, with data collection completing for three subjects, in progress for two subjects, and remaining for one subject. Due to data privacy concerns for an ongoing study, results reported are currently limited. Data collection has included 59 observations, 23 narrated tasks, 21 walk-through videos, 11 questionnaires, and 11 conferences. The observations submitted to date have consisted of all three primary reporting methods of text, audio, and video. While all three reporting methods have been regularly used, it should be noted that most participants seemed to prefer using one method over the others, and this preference varied across participants. For observation activities, the participants are instructed to select a habitability and human factors observation of their choice to report. They are free to follow the prompt provided or select a different topic. To date, the participants have elected to report on a wide variety of topics, with many observations encompassing more than one topic. While iSHORT provides tags intended to serve as memory joggers, there is no feedback regarding whether the participants have taken advantage of the tags for this purpose. Based on analog testing, while some users find the tags to be helpful, others do not use them as memory joggers. Narrated task videos are scheduled with instructions for the participant to provide a “talk aloud” narrative while completing a task already on the timeline. The task scheduled for each video is specified in the crew schedule. Note that

Task Progress:

some tasks and associated videos were performed by multiple participants (at different times) while some were performed by only one participant. The tasks captured to date include:

• Food preparation activities

• Exercise equipment setup and use

• Various payloads setup

• Blood collection process

• Housekeeping activities

• Mouse Habitat Unit removal

• Emergency OBT sim drill

• Dragon Transfer ops

• Acoustic blankets installation

Walk-through videos are scheduled with instructions for the participant to provide a tour of an area or task for human factors and habitability experts. The topic for each video was provided in the crew schedule. The areas and tasks captured to date are:

• Hygiene areas

• USOS Radiation Shelters

• Translation path from Cupola to PMM

• MARES deployed in Columbus

• US Airlock

• Robotic Work Station in Cupola

• WHC tasks

• Dining area

• Windows

• Crew Quarters

Participants complete three habitability and human factors questionnaires, corresponding to “early”, “middle”, and “late” phases of their missions. To date, three participants have finished the study and completed three questionnaires each. One participant’s data collection is ongoing and two questionnaires corresponding to the “early” and “middle” phases have been completed. The questionnaire was administered via iSHORT and iQ&A and consisted of two sections: 1) a list of 14 volume-driving mission tasks for long-duration spaceflight with check boxes to select 8 possible habitability and human factors, and 2) a Likert scale to rate the overall vehicle habitability factors in terms of privacy, habitable volume, and layout.

For the list of volume-driving tasks, participants were instructed to check a box for any of the factors that NEEDED IMPROVEMENT from a habitability point of view, or check "N/A" if none of these factors needed improvement. Each task presented the same list of factors.

Upon receiving results of the Habitability and Human Factors Questionnaire, the investigative team convenes to tailor a set of questions for the Principal Investigator (PI) Conference. The base list of questions is expanded upon and customized to reflect specific topics addressed by the participant prior to the PI Conference through observations, narrated task video, walk-through video, and the questionnaire. Tailored questions include requests for more details pertaining to walk-through and narrated task videos, observations, and problem areas indicated in the questionnaire. The participants are scheduled to set up a camera to enable a live video feed to ground. The investigative team communicates via voice at the JSC Telescience Center (TSC) with the assistance of the International Space Station Medical Project (ISSMP) support team. The PI reads the question aloud to the participant, who answers verbally. The conferences are documented by a stenographer for transcription.

Based on the data collected to date, the participants have provided thoroughly detailed and insightful feedback for all data collection types. The participants seems to have a good understanding of the types of details that will be of interest to habitability and human factors experts, and have provided a wealth of information that will be relevant for the design of future space vehicles and habitats. The investigative team anticipates that if data quality continues to be this high, this study will be highly instrumental in characterizing habitability aboard ISS and feeding into future standards and requirements. The opportunity to capture feedback about details such as privacy levels, task volumes, vehicle habitability systems, etc. throughout the course of a mission is unique and is expected to be of great value to the Space Human Factors and Habitability team.

Based on feedback from the ISS Flight Crew Integration team member who serves as a co-investigator for this study, the data gathered to date compares highly favorably to the type of information traditionally gathered in post-mission debriefs. Providing an opportunity for participants to capture observations while on-orbit allows them to give demonstrations and discuss details that are fresh on their mind. Members of the ISS operations community rely on post-mission debriefs for details such as hardware interface issues and reasoning for space utilization preferences, so capturing this information on-orbit could potentially increase awareness of operational issues, enabling the operations community to optimize crew performance, and save crew time by reducing post-mission debriefs. Following the completion of the first participant’s data collection, the study team received permission from the participant to release his data for operations and research purposes, enabling the Flight Crew Integration team to provide him with specific examples from iSHORT feedback to refer to during the Human Factors post-flight debrief. The participant provided positive feedback, indicating that having this information available was a useful enhancement to the debrief process.

	ED. NOTE (6/26/2017): Dr. Sherry Thaxton is no longer Principal Investigator as of Feb 6, 2017. New Principal Investigator will be: Maya Greene, Ph.D. (KBRwyle). Project continues with M. Greene as PI. See that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 06/01/2017)
Abstracts for Journals and Proceedings	Thaxton S, Archer R, Whitmire A, Schuh S, Vasser K. "ISS Habitability Study: Preliminary Results." Presented at 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, Feb 8 - 11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, Feb 8 - 11, 2016. , Feb-2016